

OTT User Retention Dashboard

1. Problem Statement

OTT streaming platforms face a major challenge in retaining users after subscription. Many users subscribe, engage briefly, and later stop using the platform, which increases churn and reduces revenue growth.

The objective of this project is to analyze user engagement, retention patterns, and churn behavior using Power BI dashboards. The dashboard helps business and product teams understand which factors influence retention and how user engagement can be improved.

2. Project Objective

The main objective of this project is to build an interactive Power BI dashboard to:

- Analyze user retention behavior
- Identify churn patterns
- Monitor user engagement
- Compare subscription performance
- Provide business recommendations for improving retention

3. Dataset Description

Dataset Name:synthetic_ott_user_activity.csv

The dataset contains synthetic OTT platform user activity data.

Dataset Columns

Column Name	Description
user_id	Unique user identifier
age	User age
gender	User gender
subscription_type	Subscription plan type
avg_watch_time_minutes	Average watch time
sessions_per_week	Weekly platform sessions
device_type	Device used by user
country	User country
churned	User churn status (0 = Retained, 1 = Churned)

4. Tools Used

- Power BI
- DAX
- Power Query

5. Data Cleaning Process

The following preprocessing steps were performed:

- Checked duplicate records
- Converted data types
- Verified churn values (0 and 1)
- Created calculated measures using DAX

6. KPI Measures

Total Users

Measures total unique users.

Retained Users

Counts users with churn value = 0.

Churned Users

Counts users with churn value = 1.

Retention Rate

Percentage of retained users.

Churn Rate

Percentage of churned users.

7. DAX Measures

Total Users

Total Users =

`DISTINCTCOUNT(synthetic_ott_user_activity[user_id])`

Retained Users

Retained Users =

`CALCULATE(
COUNT(synthetic_ott_user_activity[user_id]),
synthetic_ott_user_activity[churned] = 0
)`

Churned Users

Churned Users =

`CALCULATE(
COUNT(synthetic_ott_user_activity[user_id]),
synthetic_ott_user_activity[churned] = 1
)`

Retention Rate

Retention Rate =

$\text{DIVIDE}([\text{Retained Users}], [\text{Total Users}]) * 100$

Churn Rate

Churn Rate =

$\text{DIVIDE}([\text{Churned Users}], [\text{Total Users}]) * 100$

8. Dashboard Pages

Page 1 – Executive Overview

Includes:

- Total Users
- Retention Rate
- Churn Rate
- Subscription Distribution
- Country-wise Users

Page 2 – User Engagement Dashboard

Includes:

- Watch Time Analysis
- Device Usage
- Sessions Analysis

- User Engagement Trends

Page 3 – Retention Dashboard

Includes:

- Retained Users
- Churned Users
- Retention Rate
- Subscription vs Churn
- Device vs Churn
- Country vs Churn
- Cohort Analysis

Page 4 – User Performance Dashboard

Includes:

- Subscription Performance
- Device Engagement
- Country Engagement
- User Behavior Analysis

9. Business Questions Answered

1. Which factors improve retention?
2. Which subscription plans perform best?
3. Which user groups have highest churn?
4. Which device types show higher engagement?
5. Which regions have low retention?

10. Insights

- Premium users showed higher retention rates.
- Users with higher watch time had lower churn.
- Mobile users showed higher engagement.
- Some countries experienced higher churn compared to others.
- Higher session frequency improved user retention.

11. Recommendations

- Improve engagement features for low-retention users.
- Promote premium subscription plans.
- Improve retention campaigns for high churn regions.
- Increase personalized recommendations.
- Improve mobile user experience.

12. Conclusion

The OTT User Retention Dashboard successfully analyzes user behavior, retention, and churn patterns using Power BI. The dashboard provides valuable business insights that help improve customer engagement and retention strategies.